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Abstract

The Hospital Management System (HMS) is a comprehensive database management system designed to streamline the operations of a hospital by managing patient records, appointments, medical history, and billing information. This system aims to enhance the efficiency of healthcare services, improve patient care, and facilitate better communication among healthcare providers. By automating various administrative tasks, the HMS reduces human error and allows medical staff to focus more on patient care.

Background

Hospitals are complex organizations that require meticulous management of various functions, including patient admissions, treatment histories, appointment scheduling, and billing processes. Traditionally, many hospitals have relied on paper-based systems or isolated software solutions that do not communicate with one another. This fragmentation can lead to errors, lost information, and a lack of coordination among healthcare providers. The advent of digital solutions has paved the way for integrated systems that can enhance operational efficiency and improve patient outcomes.

Introduction

In today's fast-paced healthcare environment, hospitals face numerous challenges in managing patient information and ensuring quality care. Traditional methods of record-keeping often lead to inefficiencies, inaccuracies, and delays in patient treatment. A robust Hospital Management System addresses these issues by providing an integrated platform for managing various aspects of hospital operations. This project outlines the development of an HMS that focuses on improving the overall management of hospital resources while ensuring data integrity and security.

Problem of the Current System

The current systems in place at many hospitals often suffer from several critical issues:

1. Inefficiency: Manual record-keeping is time-consuming and prone to human error.

2. Data Fragmentation: Information is often stored in silos, making it difficult for healthcare professionals to access comprehensive patient data.

3. Poor Communication: Lack of integration between departments can lead to delays in treatment and misunderstandings among staff.

4. Security Risks: Paper-based systems are vulnerable to loss or damage, while existing digital systems may not have adequate security measures in place to protect sensitive patient information.

Scope of the Project

The scope of this project includes the design and implementation of a Hospital Management System that encompasses:

• Patient registration and management

• Appointment scheduling

• Medical history tracking

• Billing and insurance processing

• Reporting and analytics

• User access control and security features

Objective

The primary objectives of this project are:

1. To develop a user-friendly interface for hospital staff and patients.

2. To create a centralized database that integrates all aspects of hospital management.

3. To enhance data security and privacy measures to protect sensitive patient information.

4. To improve communication and coordination among healthcare providers.

5. To generate reports for administrative decision-making and compliance purpose

Document analysis

A Hospital Management System (HMS) using a Database Management System (DBMS) involves creating an efficient and organized system to handle various hospital functions, such as managing patient records, appointments, medical history, and billing information.

1. Requirements Analysis

Before diving into the database design, it’s essential to identify the requirements:

• Patient Management: Store patient details, including personal information, medical history, and contact information.

• Appointment Scheduling: Manage appointments between patients and healthcare providers.

• Medical Records: Track patient diagnoses, treatments, prescriptions, and allergies.

• Billing Information: Handle billing for services rendered, including insurance details and payment status.

• Staff Management: Manage information about doctors, nurses,

Database design of the system

Entity and attributes

Hospital management system have many entities and attributes. These basic entities are:

1.PATIENT(strong entity)

2.APPOINTMENT (strong entity)

3.MEDICAL HISTORY(weak entity)

4.BILLING(strong entity)

5.STAFF(strong entity)

6.Doctor (strong entity)

These each entities have their own attributes.these attributes of entities are:

1.PATIENT

✓ PatientID(key attribute)

✓ Name(composite attribute)

✓ Age(single value attribute)

✓ Gender(single value attribute)

✓ Adress(composite value attribute)

✓ Contact number(multi value attribute)

✓ Email(multi value attribute)

✓ Emergency contact(multi value attribute)

2.APPOINTMENT

✓ AppointmentID(key attribute)

✓ patientID(foreign key)

✓ DoctorID(foreign key)

✓ Date(single value attribute)

✓ Time(single value attribute)

✓ Reason(single value attribute)

3.BILLING

✓ BiillID(key attribute)

✓ patientID(foreign key)

✓ payment date(single value attribute)

✓ Total amount (single value attribute)

✓ Insurance coverage(single value attribute)

✓ Payment status(derived attribute)

4.MEDICAL HISTORY

✓ HistoryID(primary key)

✓ patientID(foreign key)

✓ DoctorID(foreign key)

✓ Diagnosis(single value attribute)

✓ Treatment(single value attribute)

✓ Prescription(single value attribute)

✓ Date(single value attribute)

5.Doctor

✓ DoctorID(primery key)

✓ Name(composit attribute)

✓ Specialization(single value attribute)

✓ phone number(multi value attribute)

✓ Availability(multi value attribute)

✓ month(single attribute)

✓ salary_in_month(single attribute)

✓salary_status(single attribute)

6.STAFF

✓ StaffID(primary key)

✓ AppointmentID(foreign key)

✓Name(composite attribute)

✓ Position(single value attribute)

✓ phone number(multi value attribute)

✓ months(single attribute)

✓ salary_in_month(single attribute)

✓salary_status(single attribute)

Relations and Cardinality

1. Patient - Appointment

Cardinality: A patient can schedule multiple appointments. there for they have one-to-many (1:N) relationship.

Participation:

Patient: Partial (some patients might not have appointments).

Appointment: Total (each appointment is associated with a patient)

2. Doctor - Appointment:

Cardinality: A doctor can attend multiple appointments. so they have one-to-many(1:N) relationship.

Participation:

Doctor: Partial (some doctors might not have appointments).

Appointment: Total (each appointment is associated with a doctor).

3. Patient - Medical History:

Cardinality: A patient's medical history consists of multiple records.they have one-to-many (1:N) relationship

Participation:

Patient: Partial (some patients might not have medical history records).

MedicalHistory: Total (each record is associated with a patient).

4. Patient - Billing:

Cardinality: A patient's billing consists of multiple records.they have one-to-many(1:N) relationship.

Participation:

Patient: Partial (some patients might not have billing records).

Billing: Total (each billing record is associated with a patient).

5. Appointment - Staff :

Cardinality: Many appointments can managed by one staff number.

They have many- to-one relationship(N:1).

Participation:

Staff: Partial (some staff might not manage appointments).

Appointment: Total (each appointment is managed by a staff member).

6.Doctor - Medical History:

Cardinality: A doctor can be associated with multiple medical history records.they have one-to-many relationship(1:N).

Participation:

Doctor: Partial (some doctors might not have medical history records).

MedicalHistory: Total (each record is associated with a doctor).

Knowing The role name between the relationship of each entities are essential for drawing ER diagram.these roll names are:

✓ Patient schedules Appointment

✓ Medical history reviewed by Doctor

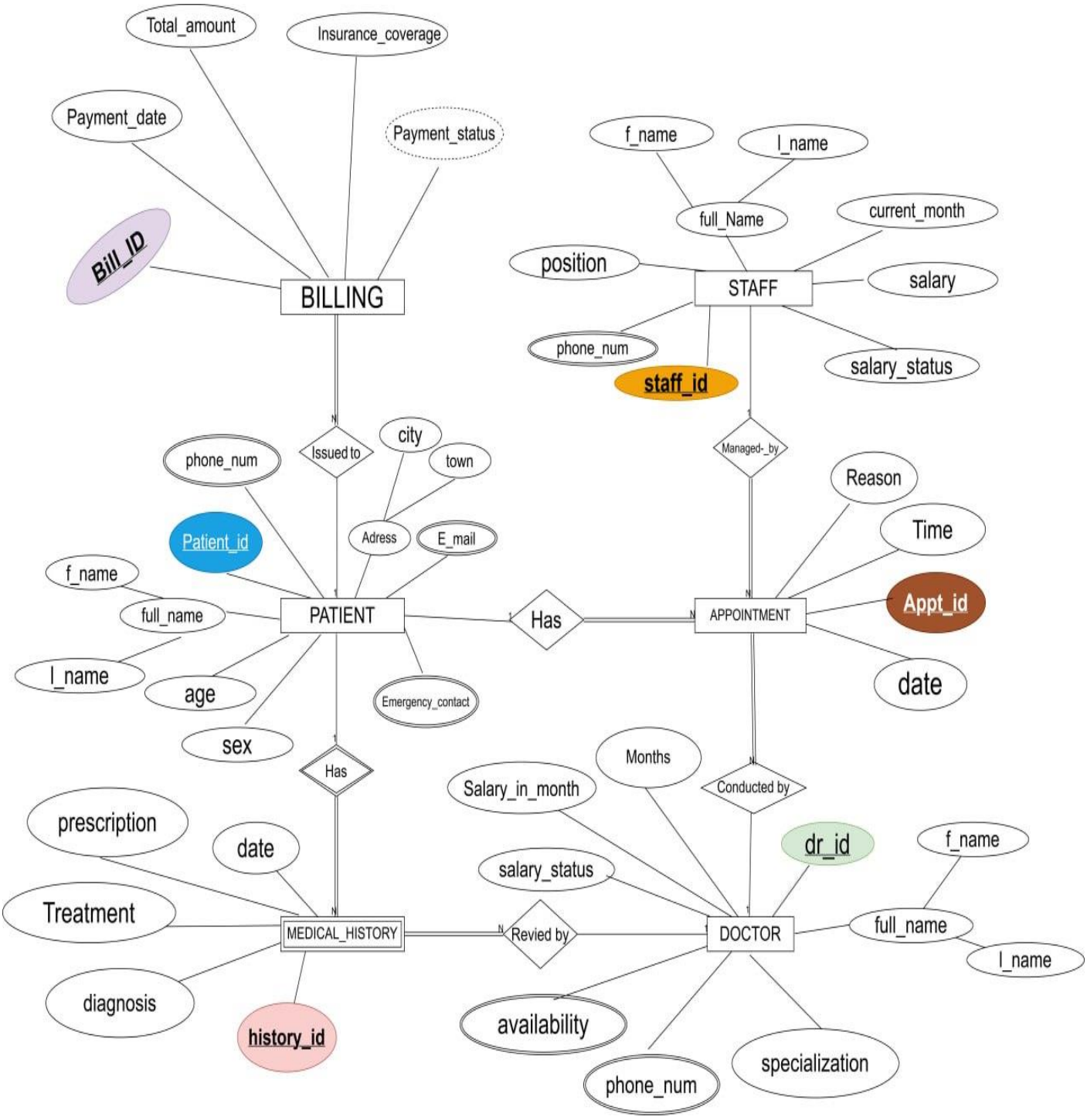
✓ Appointments managed by Staff

✓ billing issued to patient

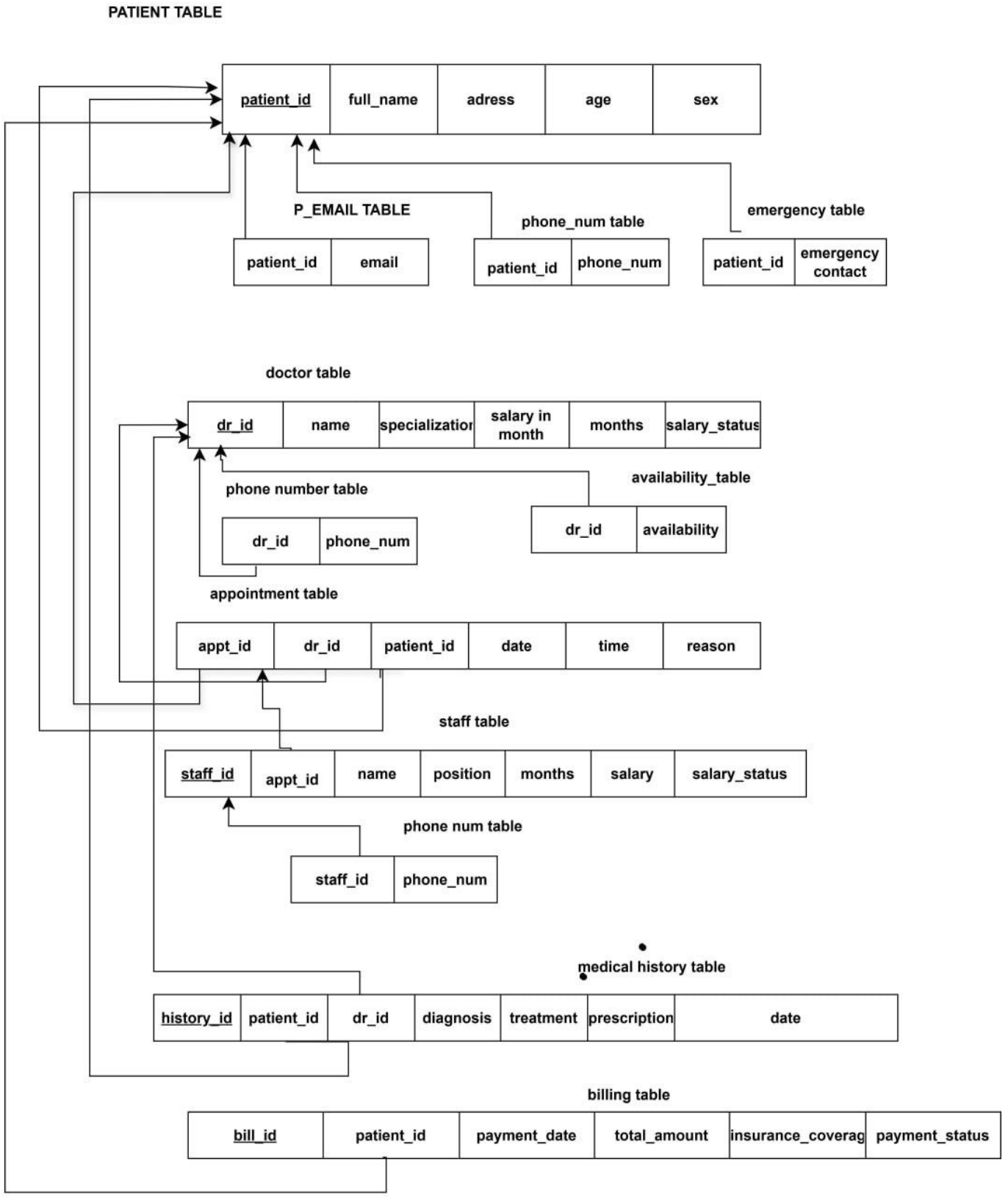
✓ Patient has Medical history

✓ Appointment conducted by doctor

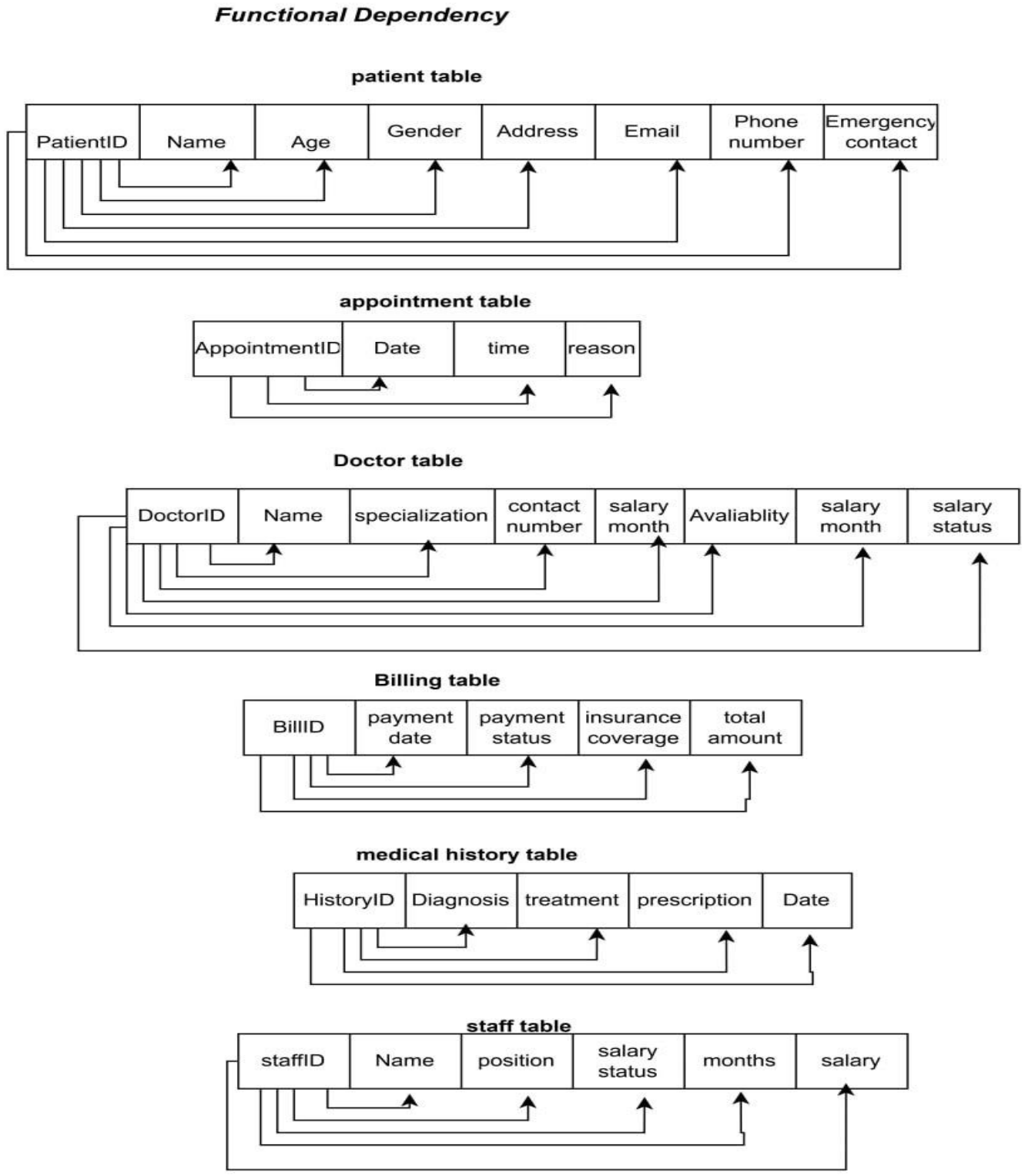
ER diagram


Total_amount
Insurance_coverage
Payment_date
Payment_status
Bill_ID
BILLING
f_name
l_name
full_Name
current_month
position
STAFF
salary
phone_num
staff_id
salary_status
Issued to
city
town
phone_num
Patient_id
Adress
E_mail
Managed-_by
Reason
Time
f_name
full_name
PATIENT
Has
APPOINTMENT
Appt_id
l_name
age
Emergency_contact
date
sex
Has
Months
Salary_in_month
Conducted by
prescription
date
dr_id
f_name
salary_status
Treatment
MEDICAL_HISTORY
Revied by
DOCTOR
full_name
l_name
diagnosis
history_id
availability
specialization
phone_num

Mapping ER diagram to relational schema



Functional dependency of the system



Normalization

First normalization

First normalization of patient

Patient_id	F_name	L_name	city	town	age	sex

(i.e full name and address is divided since they are composite attribute)

Example:

Patient_id	Full_name	Adress	age	sex
1	Mohammed Hassen	DESSIE GIRANA	23	MALE

First normalization will be:

Patient_id	F_name	L_name	city	town	age	sex
1	Mohammed	Hassen	Dessie	Girana	23	Male

And also phone num, emergency contact, email are normalize in the first normal form.

Patient_id	Phone_num
1	090909099 092343245

Patient_id	emergency_contact
1	098766442 0987654433

Patient_id	email
1	compliment123@gmail.com moh124@gmail.com

First normal form will be as follows respectively:

Patient_id	Phone_num
1	090909099
1	092343245

Patient_id	emergency_contact
1	098766442
1	0987654433

Patient_id	email
1	compliment123@gmail.com
1	moh124@gmail.com

First normalization of doctor table:

In this table names are divided because it is composite attribute the table after first normalization will be:

Dr_id	F_name	L_name	Specialization	Salary in month	month	Salary_status

Phone number and availability also multi valued attributes in this case the should normalize in first norm:

If the doctor has multiple availability and multiple phone number it should it should normalize as patients phone number and other attributes. Example

Dr_id	availability
2	Monday morning Friday afternoon

Dr_id	Phone_num
2	07776567676 0975875

The first normalize form of tables are as follows :

Dr_id	availability
2	Monday morning
2	Friday afternoon

Dr_id	Phone_num
2	07776567676
2	0975875

first normalization of staff table :

in this table also name is divided because it is composite attribute, phone number also divided because it is multi valued attribute

Staff_id	Appt_id	F_name	L_name	Position	Salary_in_month	Current_month	Salary_status

Phone number table is normalize as follows:

Before normalization:

Staff_id	Phone_num
63243	0987652 07608765478

After normalization:

Staff_id	Phone_num
63243	0987652
63243	07608765478

Appointment, medical history, and billing tables need not first normal form because in first normal form already.

Second normalization

All tables are in second normal therefore they need not second normal form.

Third normalization

Third normalization is needed in billing table because payment_status is derived from payment date so it should remove by the rule of third normal form.

Before normalization:

Bill_id	Payment_date	Total_amount	Insurance_coverage	Payment_status
5465	01/03/2017 E.C	25454BIRR	230	Paid

After normalization:

Bill_id	Payment_date	Total_amount	Insurance_coverage
5465	01/03/2017 E.C	25454BIRR	230

Database implementation on SQL command

```
create database hospitalmanagementsystem11;
CREATE TABLE Patient (
   PatientID INT PRIMARY KEY identity(1,1),
   FirstName VARCHAR(50),
   LastName VARCHAR(50),
age int CHECK(age>0),
   sex varchar(6),
   Phone int ,
        emergency_contact int,
email varchar(100),
   Address varchar(50));
Insert into patient
values('Abebe','kebede',23,'male',0920253476,12345,'abb@gemail.com','woldia');
Insert into patient
values('abdu','ahmed',23,'male',098765432,1234500,'aaa@gemail.com','girana');
Insert into patient
values('oumer','awol',22,'male',0920252222,123045,'123o@gemail.com','kutaber');

CREATE TABLE Doctor ( DoctorID INT PRIMARY KEY identity (5,1), FirstName
VARCHAR(30) NOT NULL,LastName varchar(30),
Specialization VARCHAR(100),  phone int,  Availability VARCHAR(50),salary_in_month
int,months varchar(30),salary_status varchar(30));
INSERT INTO Doctor values('Abrham','worku','cardiology',0920335675,'monday
morning',20000,'september','paid');
INSERT INTO Doctor values('osman','yimam','dental',0920335335,'monday
afternoon',40000,'october','paid');
INSERT INTO Doctor values('muhammed','worku','cardiology',0920035675,'friday
morning',10000,'jun','paid');



CREATE TABLE Appointments (AppointmentID INT PRIMARY KEY identity(9,1),PatientID
INT foreign key references patient(patientid),
   DoctorID INT foreign key references doctor(doctorid),
   AppointmentDate varchar(30),
Appointmenttime varchar(30),
   reason varchar(50))

Insert into appointments values(1,5,'2016-03-17','04:30:00','headache treatment');
Insert into appointments values(2,6,'2023-09-18','05:30:00','knee pain consulatin');
Insert into appointments values(3,7,'2024-03-16','08:00:00','routin check up');

CREATE TABLE MedicalHistory (HistoryID INT PRIMARY KEY identity (10,1), PatientID
INT foreign key references patient(patientid),
```

```
DoctorID int foreign key references doctor(doctorid),Diagnosis VARCHAR(255),
Treatment VARCHAR(255),
 Prescription VARCHAR(255),  Date varchar(30))

INSERT INTO Medicalhistory values(1,5,'hyper tention','life style chang','Amlodipine
500g','2016-04-22');
INSERT INTO Medicalhistory values(2,6,'migraine','medication prescribed','sumatriptan
50g','2016-02-27');
INSERT INTO Medicalhistory values (3,7,'flue','rest and hydration','oseltamivir 50g','2016-03-
24');

CREATE TABLE Billing ( BillID INT PRIMARY KEY identity (20,1),  PatientID INT foreign
key references patient(patientid), TotalAmount int,
InsuranceCoverage int,PaymentStatus VARCHAR(20),  PaymentDate varchar(10))

Insert into billing values
(1, 150, 50, 'Paid', '2016-02-01');
Insert into billing values(2,5000,1500,'unpaid','2016-03');
Insert into billing values(3,1000,500, 'approved','2016-04-18');

 CREATE TABLE Staff ( StaffID INT PRIMARY KEY identity(6,1),AppointmentID INT
foreign key references appointments(appointmentid),
 First_name VARCHAR(30) NOT NULL,Lastname varchar(30),
Position VARCHAR(50), phone int, Salary int,months varchar(20),salaryStatus varchar(20))
INSERT INTO Staff VALUES(9,'Natnael','Abera','nurse','0915345234',7000,'june','paid');
INSERT INTO Staff VALUES(10,'osman','Abera','nurse','0911115234',7700,'october','paid');
INSERT INTO Staff VALUES(11,'Natnael','Abera','nurse','0915322234',7000,'june','paid');
```

Conclusion

The development of a Hospital Management System presents an opportunity to revolutionize how hospitals manage their operations. By addressing the shortcomings of current systems and implementing a comprehensive digital solution, hospitals can improve patient care, streamline processes, and enhance overall efficiency. Through careful planning, stakeholder engagement, and rigorous testing, this project aims to deliver a robust HMS that meets the needs of healthcare providers and patients alike.